

L8 Other notions of equilibria

CS 280 Algorithmic Game Theory

Ioannis Panageas

Relaxing Nash equilibrium

- NASH is computationally hard.

Question: Are there other equilibrium notions that are computationally tractable?

Relaxing Nash equilibrium

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Answer: Correlated equilibria, i.e., relaxing the product distribution assumption.

Example (Correlated eq.)

	Chicken-out	Dare
Chicken-out	0, 0	-2, 1
Dare	1, -2	-10, -10

Suppose agents are **recommended** (C, D) , (D, C) , (C, C) with probability $\frac{1}{3}$ each.

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Suppose agents are **recommended** (C, D) , (D, C) , (C, C) with probability $\frac{1}{3}$ each.

- If agent row is **recommended to choose C** , then column is recommended to play **C or D** with equal probability. Expected payoff of row is $\frac{1}{2} \cdot 0 + \frac{1}{2}(-2) = -1$ which is **greater than switching to D** (expected payoff is -4.5).

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- If agent row is **recommended to choose C** , then column is recommended to play **C or D** with equal probability. Expected payoff of row is $\frac{1}{2} \cdot 0 + \frac{1}{2}(-2) = -1$ which is **greater than switching to D** (expected payoff is -4.5).
- If agent row is **recommended to choose D** , then column is recommended to play **C** . Expected payoff of row is 1 which is **greater than switching to C** (expected payoff is 0).

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Suppose agents are **recommended** (C, D) , (D, C) , (C, C) with probability $\frac{1}{3}$ each.

- If agent row is **recommended to choose C** , then column is recommended to play **C or D with equal probability**. Expected payoff of row is $\frac{1}{2} \cdot 0 + \frac{1}{2}(-2) = -1$ while column's expected payoff is -4.5).
- If agent row is **(C, D) , (D, C) and (C, C) with probability $\frac{1}{3}$ each** is recommended to play **C** . Expected payoff of row is $\frac{1}{3} \cdot 0 + \frac{1}{3}(-2) + \frac{1}{3}(1) = -\frac{1}{3}$ while column's expected payoff is $\frac{1}{3} \cdot 1 + \frac{1}{3}(-2) + \frac{1}{3}(-10) = -\frac{11}{3}$. **than switching to C** (expected payoff is $-\frac{11}{3}$).

Similarly for column player!

(C, D) , (D, C) and (C, C) with probability $\frac{1}{3}$ each is a correlated eq.

Definitions

Definition (Recall). *A game is specified by*

- *set of n players $[n] = \{1, \dots, n\}$*
- *For each player i a set of strategies/actions S_i .*
- *set of strategy profiles $S = S_1 \times \dots \times S_n$.*
- *Each agent i has a utility $u_i : S \rightarrow [-1, 1]$ denoting the payoff of i .*

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Definition (Correlated Equilibrium). *Correlated equilibrium is a **distribution** χ over S such that for all agents i and strategies b, b' of i*

$$\mathbb{E}_{s \sim \chi}[u_i(b, s_{-i}) | s_i = b] \geq \mathbb{E}_{s \sim \chi}[u_i(b', s_{-i}) | s_i = b].$$

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Similarly for all agents i and swapping functions $f : S_i \rightarrow S_i$,

$$\mathbb{E}_{s \sim \chi}[u_i(s_i, s_{-i})] \geq \mathbb{E}_{s \sim \chi}[u_i(f(s_i), s_{-i})].$$

Correlated equilibrium and Nash

Remarks:

- Knowing an agent her recommended action, she can **infer** something about other players' moves. Yet she is **better off playing** the **recommended** action.
- Suppose χ is a **product distribution**. Then correlated eq. corresponds to Nash eq.

Correlated equilibrium and Nash

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- Knowing an agent her recommended action, she can **infer** something about other players' moves. Yet she is **better off playing** the **recommended** action.

Set of Nash equilibria $\subseteq$ Set of correlated equilibria.

Example (Coarse Correlated eq.)

			
	0, 0	-1, 1	1, -1
	1, -1	0, 0	-1, 1
	-1, 1	1, -1	0, 0

Suppose the actions (R, P) , (R, S) , (P, R) , (P, S) , (S, R) , (S, P) are chosen with probability $\frac{1}{6}$ each.

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- If agent row plays R , agent column responds with either P or S with equal probability. If column deviates (say starts responding with paper higher possibility) she will incur more loss when row plays S (recall row plays R as well S with equal probability).

Example (Coarse Correlated eq.)

			
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- If agent column is instructed to play P then she knows that **other player is playing either R or S** and column has **average payoff 0**. She can change then to R and **improve payoff to $1/2$** compared to zero if she plays recommended action. In this case, column could exploit knowledge of action recommendation to improve her payoff.

Definitions

Definition (Coarse Correlated Equilibrium). *Coarse correlated equilibrium is a **distribution** χ over S such that for all agents i and strategies b of i*

$$\mathbb{E}_{s \sim \chi}[u_i(s)] \geq \mathbb{E}_{s \sim \chi}[u_i(b, s_{-i})].$$

Remark: The difference between coarse correlated and correlated is that we can choose a “smart” **swap function**, namely f “knows” the distribution χ .

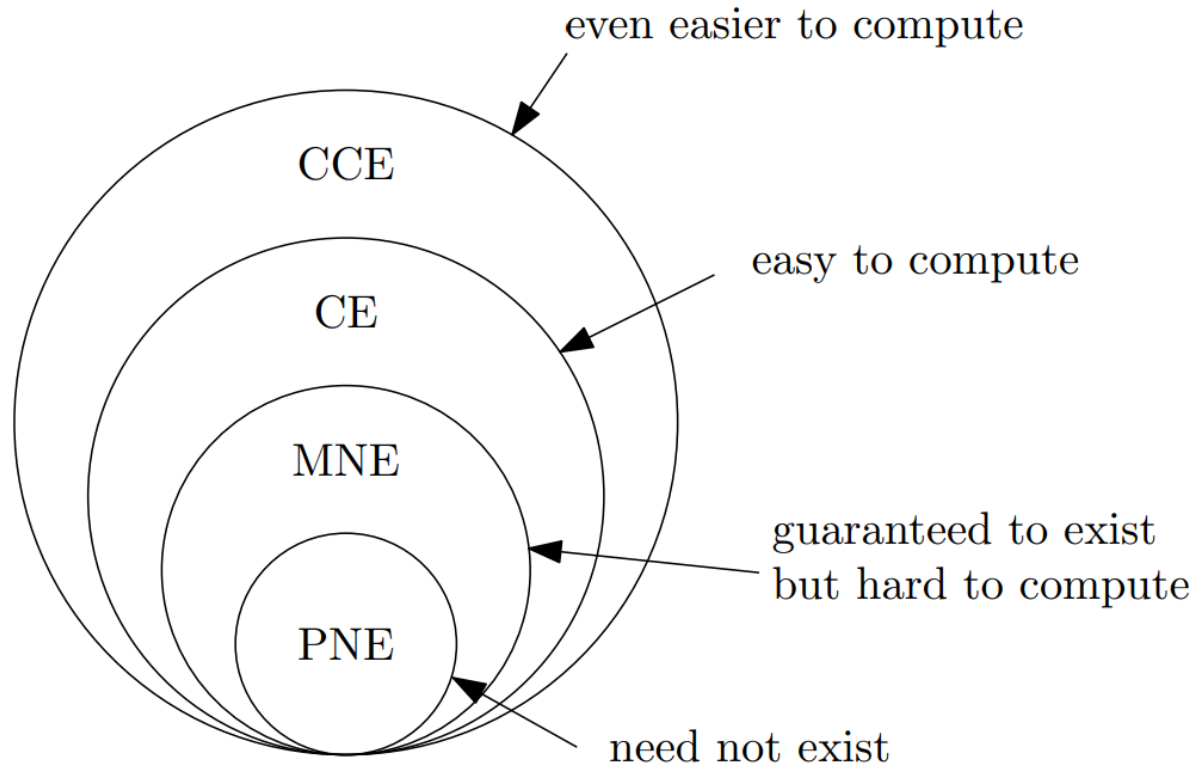
Definitions

Definition (Coarse Correlated Equilibrium). Coarse correlated equilibrium is a *distribution* χ over S such that for all agents i and strategies b of i

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Set of correlated equilibria $\subseteq$ Set of coarse correlated equilibria.

Full picture of Inclusions



Online learning in Games

Definition. *At each time step $t = 1 \dots T$.*

- *Each **player** i chooses $x_i^{(t)} \in \Delta_i$ (simplex).*
- *Each **player** experiences payoff $u_i(x^{(t)})$ and observes all players strategies $x_j^{(t)}$.*

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- Each *player i* chooses $x_i^{(t)} \in \Delta_i$ (simplex).
- Each *player* experiences payoff $u_i(x^{(t)})$ and observes all players strategies $x_j^{(t)}$.

Player's i goal is to minimize the (time average) **Regret**, that is:

$$\frac{1}{T} \left[\max_{a \in S_i} \sum_{t=1}^T u_i(a, x_{-i}^{(t)}) - \sum_{t=1}^T u_i(x^{(t)}) \right].$$

If $\text{Regret} \rightarrow 0$ as $T \rightarrow \infty$, the algorithm is called **no-regret**.

A no-regret Algorithm

Definition (Online Gradient Descent). Let $\ell_t : \mathcal{X} \rightarrow \mathbb{R}$ be family of convex functions, differentiable and L -Lipschitz in some compact convex set $\mathcal{X}$ of diameter D . Online GD is defined:

Initialize at some x_0 .

For $t:=1$ to T do

1. $y_t = x_t - \alpha_t \nabla \ell_t(x_t)$.

2. $x_{t+1} = \Pi_{\mathcal{X}}(y_t)$.

Regret: $\frac{1}{T} \left(\sum_{t=1}^T \ell_t(x_t) - \min_x \sum_{t=1}^T \ell_t(x) \right)$.

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step-size

$\ell_t = -u_i(x^{(t)})$

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Analysis of Online GD for L -Lipschitz

Theorem (Online Gradient Descent). Let $\ell_t : \mathcal{X} \rightarrow \mathbb{R}$ be family of convex functions, differentiable and L -Lipschitz in some compact convex set $\mathcal{X}$ of diameter D . It holds

$$\left(\frac{1}{T} \sum_{t=1}^T \ell_t(x_t) - \min_x \sum_{t=1}^T \ell_t(x) \right) \leq \frac{3}{2} \frac{LD}{\sqrt{T}},$$

with appropriately choosing $\alpha = \frac{D}{L\sqrt{t}}$.

Remarks:

- If we want error ϵ , we need $T = \Theta\left(\frac{L^2 D^2}{\epsilon^2}\right)$ iterations.
- I could have written **Multiplicative Weights Update**. This is **another** no-regret algorithm! Same regret guarantees, i.e., $O\left(\frac{1}{\sqrt{T}}\right)$.

Analysis of Online GD for L -Lipschitz

Proof. Let x^* be the argmin of $\sum \ell_t(x)$.

$$\begin{aligned}\ell_t(x_t) - \ell_t(x^*) &\leq \nabla \ell_t(x_t)^\top (x_t - x^*) \text{ convexity,} \\ &= \frac{1}{\alpha_t} (x_t - y_t)^\top (x_t - x^*) \text{ definition of GD,}\end{aligned}$$

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Analysis of Online GD for L -Lipschitz

Proof cont. Since

$$\ell_t(x_t) - \ell_t(x^*) \leq \frac{1}{2\alpha_t} \left(\|x_t - x^*\|_2^2 - \|x_{t+1} - x^*\|_2^2 \right) + \frac{\alpha_t L^2}{2},$$

taking the telescopic sum we have

$$\begin{aligned} \sum_{t=1}^T (\ell_t(x_t) - \ell_t(x^*)) &\leq \sum_{t=1}^T \|x_t - x^*\|_2^2 \left(\frac{1}{2\alpha_t} - \frac{1}{2\alpha_{t-1}} \right) + \frac{L^2}{2} \sum_{t=1}^T \alpha_t. \\ &\leq \frac{D^2}{2} \sum_{t=1}^T \left(\frac{1}{\alpha_t} - \frac{1}{\alpha_{t-1}} \right) + \frac{L^2}{2} \sum_{t=1}^T \alpha_t. \end{aligned}$$

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where we used the fact $\sum \frac{1}{\sqrt{t}} \leq 2\sqrt{T}$ and $\alpha_t = \frac{D}{\sqrt{t}L}$.

Computing coarse correlated equilibria

Suppose that each agent i uses **no-regret dynamics (online GD)**, with $l_t = -u_i(x^{(t)})$ where $x^{(t)}$ is the mixed strategy profile at **iterate t** .

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We conclude that for each agent i

$$\mathbb{E}_{s \sim \sigma}[u_i(s)] = \frac{1}{T} \sum_{t=1}^T \mathbb{E}_{s \sim \sigma^t}[u_i(s)]$$
$$\min_{b \in S_i} \mathbb{E}_{s \sim \sigma}[u_i(b, s_{-i})] = \min_{b \in S_i} \frac{1}{T} \sum_{t=1}^T \mathbb{E}_{s \sim \sigma^t}[u_i(b, s_{-i})]$$

Computing coarse correlated equilibria

Suppose
where σ

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We can

$$\min_{b \in S_i} \mathbb{E}_{s \sim \sigma} [u_i(b, s_{-i})] - \mathbb{E}_{s \sim \sigma} [u_i(s)] \leq \frac{3}{2} \frac{\sqrt{n}}{\sqrt{T}}$$

Lipschitz

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Choosing $T = \frac{9n}{4\epsilon^2}$ we conclude σ is ϵ -approximate CCE!

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We can

If we use MWUA, it gives $O\left(\frac{\ln n}{\epsilon^2}\right)$.

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